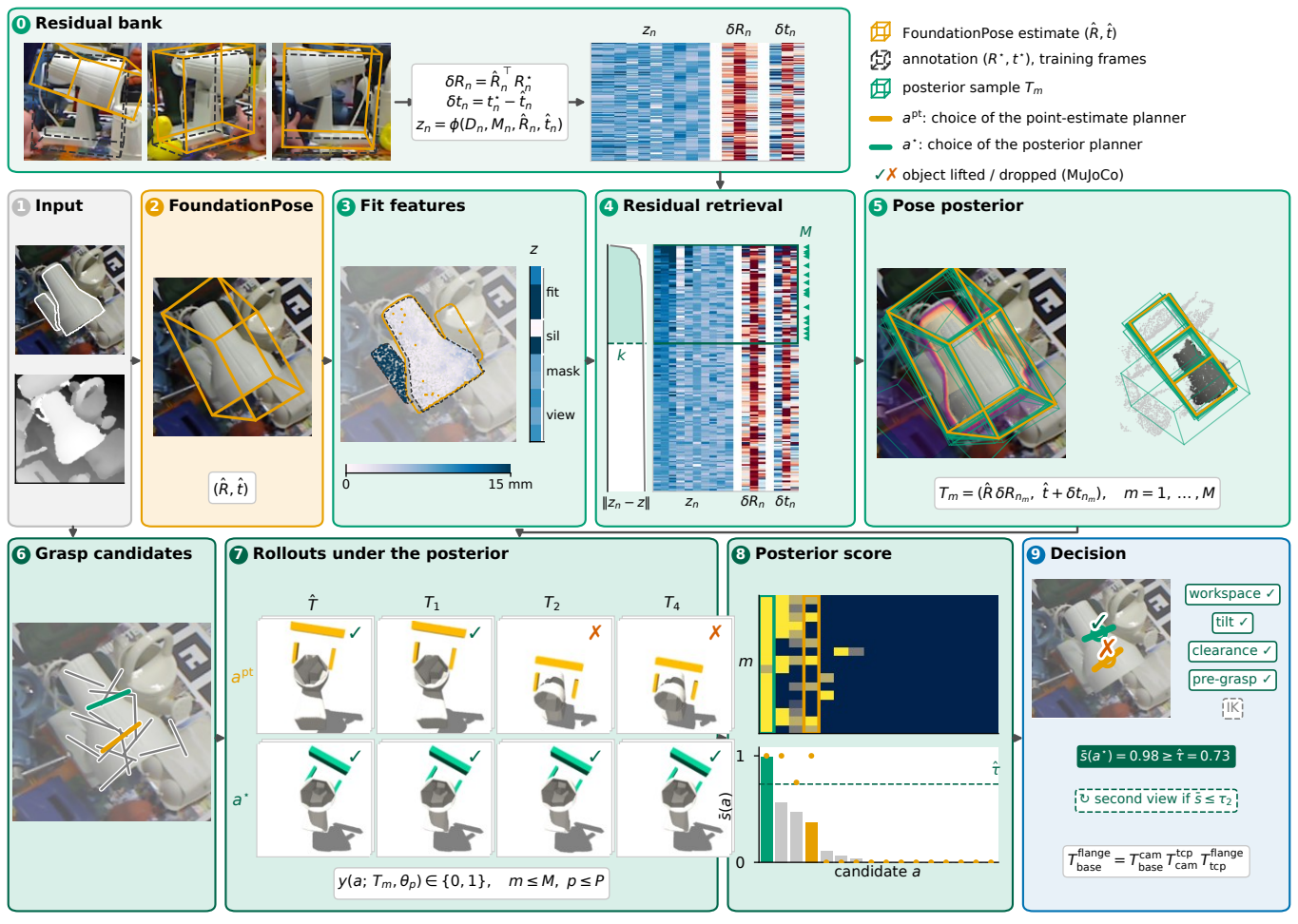

0 Residual bank
$\delta R_n = \hat{R}_n^\top R_n^\star$
$\delta t_n = t_n^\star - \hat{t}_n$
$z_n = \phi(D_n, M_n, \hat{R}_n, \hat{t}_n)$
z_n
δR_n
δt_n
FoundationPose estimate $(\hat{R}, \hat{t})$
annotation $(R^\star, t^\star)$, training frames
posterior sample T_m
a^{pt}: choice of the point-estimate planner
$a^\star$: choice of the posterior planner
✓✗ object lifted / dropped (MuJoCo)
1 Input
2 FoundationPose
$(\hat{R}, \hat{t})$
3 Fit features
z
fit
sil
mask
view
0
15 mm
4 Residual retrieval
M
k
$\|z_n - z\|$
z_n
δR_n
δt_n
5 Pose posterior
$T_m = (\hat{R}\,\delta R_{n_m},\ \hat{t} + \delta t_{n_m}), \quad m = 1, \ldots, M$
6 Grasp candidates
7 Rollouts under the posterior
$\hat{T}$
T_1
T_2
T_4
a^{pt}
$a^\star$
$y(a; T_m, \theta_p) \in \{0, 1\}, \quad m \le M,\ p \le P$
8 Posterior score
m
1
0
$\bar{s}(a)$
$\hat{\tau}$
candidate a
9 Decision
workspace ✓
tilt ✓
clearance ✓
pre-grasp ✓
IK
$\bar{s}(a^\star) = 0.98 \ge \hat{\tau} = 0.73$
↻ second view if $\bar{s} \le \tau_2$
$T_{\text{base}}^{\text{flange}} = T_{\text{base}}^{\text{cam}}\, T_{\text{cam}}^{\text{tcp}}\, T_{\text{tcp}}^{\text{flange}}$